ADD: RONGGUI STREET, SHUNDE DISTRICT, FOSHAN
GUANGDONG PROVINCE, 528300, CHINA
PHONE: +852-84919320 • E-MAIL: HUANGHANQI1123@163.COM
WEB: HANSON5732.GITHUB.IO • GITHUB: GITHUB.COM/HANSON5732

HUANG HANQI

EDUCATION

08/2026-07/2027 University of Hong Kong
Master of Science in Computer Science (Financial Computing Stream)

08/2022-07/2026 Macau Polytechnic University
Bachelor of Science in Computing

- Final CGPA 3.67/4, ranked Top 5 in the programme and awarded the 2024-25 Macao Fok Ying Tung Foundation Scholarship
- Huawei ICT Competition (Macao), Cloud Track Excellence Award; summer exchange at the UM-SJTU Joint Institute, Shanghai Jiao Tong University (2023)

RESEARCH

08/2025-04/2026 Electronic Properties of Functional Materials
Final-year project: high-throughput DFT and machine learning

- Automated a DFT pipeline (VASP, Pymatgen) on an ArchLinux supercomputer, curating 2,598 structures and 2,552 formation energies
- Implemented CGCNN in PyTorch and compared it with XGBoost and Random Forest: the graph network was most accurate on electronic properties (Fermi level R^2 0.9635, band gap R^2 0.9584) while XGBoost led on formation energy (R^2 0.986)
- Feature-importance analysis linked electronic properties to lattice topology and stability to local orbitals, which guided model selection

03/2025-08/2025 HypoSelectSimRAG: Multi-Path Self-Consistent Query Translation
Second author, IEEE ICCEIC 2025

- Proposed a RAG framework that generates hypothetical documents along four temperature-guided paths and keeps only the candidate closest to the query
- Implemented it with LangChain and compared five document-selection strategies through ablations on the 1,200-sample rag-dataset-1200 benchmark
- Raised Factual Correctness to 0.6910 under RAGAS, against 0.5257 for standard RAG and 0.6178 for HyDE
- Published at the 6th International Conference on Computer Engineering and Intelligent Control, indexed in IEEE Xplore, DOI 10.1109/ICCEIC67916.2025.11309165

INTERNSHIPS

07/2026-08/2026 Full-Stack Engineer, Development Department, Shunde Rural Commercial Bank, Foshan, China

- ❖ Delivered the bank's green-finance whitelist module end to end, from REST APIs on the internal service stack to the React management page, and released it through code review and deployment
- ❖ Replaced Apache POI with EasyExcel streaming read and write to eliminate memory overflow on hundred-megabyte client spreadsheets
- ❖ Analysed slow SQL and added composite indexes across the three tables joined by whitelist verification, reducing check latency from seconds to milliseconds
- ❖ Scheduled Spring Task jobs to parse Excel files from Alibaba Cloud OSS and persist them automatically, with retry handling and audit logging

06/2025-08/2025 Data Analyst & Back-end Engineer, Development Department, Foshan Dayan Data Technology Co. Ltd., Foshan, China

- ❖ Independently built the order management module of a glass factory's order preparation system, cutting preparation time from 8 hours to 1 hour (87.5%)
- ❖ Chose a pdfplumber-first pipeline with OCR fallback and LLM structured extraction over a trained computer-vision model, and handled the PDF formats of 8 manufacturers
- ❖ Built the glass-name generation workflow on a private Dify deployment with the DeepSeek API, using few-shot and chain-of-thought prompts and a domain knowledge base
- ❖ Helped build a retrieval-augmented generation model for an AI email reply application, whose response accuracy reached 88%

06/2024-07/2024 Cybersecurity Intern, Data Administration Division, Shunde District Government Service and Data Administration Bureau, Foshan, China

- ❖ Joined the city's cybersecurity drill, remediating website vulnerabilities with Burp Suite, Kali Linux and SQLmap, and wrote the assessment reports
- ❖ Secured the daily protection of the bureau by analysing network flows, building HFish honeypots and configuring iptables on Linux

24/07/2023-19/08/2023 Back-End Engineer, Data Administration Division, Shunde Rural Commercial Bank, Foshan, China

- ❖ Assisted in developing the bank's internal personnel management system with Spring Cloud and MySQL, and took part in its API testing and deployment

OTHERS

Language Proficiency: Native in Cantonese and Mandarin, fluent in English (IELTS overall 7.0)
Computer Literacy: LangChain, Dify, LangSmith and RAGAS; Python (Django, Flask, FastAPI, NumPy, Pandas, PyTorch), Java (Spring Boot), JavaScript (Vue.js, React); MySQL, Redis, Nginx and Linux